

Open Economy Carbon Pricing: A Trade-Based Framework for Efficient, Politically Feasible, and Rapid Decarbonisation

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Abstract

Carbon pricing remains too weak in most countries to deliver rapid decarbonisation. We argue this weakness is structural, arising from three constraints: (1) the trade-exposure constraint, (2) the political-economy constraint, and (3) the primary-to-quaternary structure of energy transition. Conventional carbon pricing (CCP) applies a source-based tax to domestic production, creating competitiveness risks and inviting a race to the bottom similar to corporate income taxation. Open Economy Carbon Pricing (OECPP) restructures carbon pricing into a destination-based framework, pairing a production-side feebate with a consumption-based carbon charge that applies symmetrically to imports and domestic goods while zero-rating exports. This enables a race to the top akin to the rise of VAT. We present the simplest integrated implementation mechanism using CBAM logic and common external tariff structures. Appendices provide a simple trade model and a simple political-feasibility model.

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1 Introduction

Carbon pricing is widely regarded as a foundational instrument for climate policy. Yet after three decades of implementation, effective carbon prices remain far below levels required for accelerated decarbonisation. The common explanation is political resistance, but this paper argues that the true problem is structural: *conventional carbon pricing (CCP) is a source-based production tax*, and this architecture embeds three deep constraints that no amount of political effort can fully overcome.

First, CCP induces a persistent *trade constraint*. Because the tax is applied where production occurs, it raises domestic production costs, weakens export competitiveness, and encourages import substitution from jurisdictions with lower carbon constraints. This creates an international race to the bottom, analogous to the erosion of corporate income tax rates.

Second, CCP suffers from a severe *political-economy constraint*. Carbon taxes and ETS compliance costs are highly salient for both producers and households: producers face higher average costs and lobby against increases, while households face higher retail prices and vote accordingly. The politically feasible carbon price is therefore limited well below the level required to induce capital turnover in primary sectors.

Third, CCP creates a *transition constraint*. Primary sectors such as steel, cement, electricity, refining, and heavy transport are capital-intensive and slow to respond to marginal price changes. A carbon price high enough to accelerate early switching is politically infeasible under CCP’s architecture.

This paper proposes a structural redesign: **Open Economy Carbon Pricing (OECF)**, which resolves all three constraints by separating the functions of carbon pricing across two instruments:

1. a **revenue-neutral production feebate** that strengthens clean–dirty competition without raising average production costs; and
2. a **destination-based consumption carbon charge** applied to all goods consumed domestically, with exports zero-rated.

This architecture mirrors the transformation that the Value Added Tax (VAT) achieved when replacing source-based turnover taxes: shifting taxation to the destination basis eliminates competitiveness harms and enables a “race to the top.”

Before developing the detailed implementation (Part II) and models (Part III), Part I sets out four foundational components:

1. the relevant literatures;

2. the structural constraints of CCP;
3. the OECP framework; and
4. the **two-plus-one functions of carbon pricing**, a new conceptual lens that clarifies why CCP fails and why OECP works.

2 Literature Review

Open Economy Carbon Pricing (OECP) sits at the intersection of climate economics, international trade, environmental taxation, and public finance. The relevant literature spans six strands: (1) competitiveness and leakage under unilateral carbon pricing; (2) benchmarked instruments, feebates, and output-based rebating; (3) border carbon adjustments and carbon-adjusted tariff structures; (4) consumption-based greenhouse gas accounting; (5) consumption-based environmental taxation; and (6) political acceptability and destination-based taxation in public finance.

This section reviews each in turn and highlights how OECP synthesises their insights.

2.1 Competitiveness, Leakage, and Unilateral Carbon Pricing

A substantial body of work has shown that conventional, source-based carbon pricing creates competitiveness concerns and carbon leakage when implemented unilaterally. Seminal contributions include Fischer and Fox (2007) and Fischer and Fox (2012), who show analytically and numerically that carbon pricing raises costs in energy-intensive, trade-exposed (EITE) sectors and shifts production abroad. General equilibrium analyses such as Böhringer and Rutherford (2016) confirm that leakage rates can be substantial even under moderate carbon prices.

Theoretical research in environmental trade theory (Hoel (1996); Copeland and Taylor (2005)) demonstrates that open economies internalise only a fraction of the environmental externality, setting carbon prices below global optima. Aldy and Pizer (2015) further show that carbon-price differentials across countries can persist even under cooperative arrangements when carbon pricing remains production-based.

This literature establishes that ****source-based carbon pricing induces a structural race to the bottom****, creating the core policy problem that OECP seeks to resolve.

2.2 Feebates, Benchmarks, and Output-Based Rebating

Benchmark-based policy instruments offer relative incentives without raising average costs. Fischer and Fox (2012) and related work (Fischer (2019)) show that output-based rebating (OBR) protects competitiveness while preserving some marginal abatement incentive. Feebate systems—analysed in Böhringer and Lange (2005) and applied in multiple sectors including vehicles, electricity, and industry (Holland et al. (2020))—create linear transfers based on emissions intensity relative to a benchmark.

The key insight is that feebates can provide robust **relative** incentives to decarbonise while avoiding **absolute** cost increases, thereby eliminating the primary source of industrial opposition to carbon pricing.

OECP generalises this approach by using a sector-wide revenue-neutral feebate as the production-side instrument, enabling high incentive rates without harming competitiveness.

2.3 Border Carbon Adjustments and Carbon-Adjusted Tariffs

Border carbon adjustments (BCAs) have been widely proposed as a response to competitiveness and leakage concerns. Foundational work by Ismer and Neuhoﬀ (2007) and Mattoo et al. (2009) examines how import charges and export rebates could mimic the effects of a domestic carbon price at the border. Subsequent work synthesising legal and economic perspectives (Mehling et al. (2019); Droege and Fischer (2020)) identifies persistent challenges: verifying embodied emissions, assessing foreign climate policies, treating free allocation, and ensuring WTO consistency.

More recent literature extends border adjustments toward **environmental tariff reforms**, often referred to as “green tariffs.” This work includes environmental border charges (Lockwood and Whalley (2008)), early proposals for climate-related tariff adjustments (Low (2008); Pauwelyn (2007)), and analyses of integrating carbon criteria into the EU’s Common External Tariff (Colares (2014)). Studies on differentiated tariffs based on verified low-carbon production (Électricité de France (2019)) and HS-level embodied-emissions adjustments (Moran and Wood (2020); Brandi (2023)) demonstrate that customs-based climate policy is legally and operationally feasible.

These contributions highlight two insights reflected in OECP: (1) customs systems can incorporate carbon content using benchmark intensities; (2) destination-based adjustments can be more transparent and administratively simpler than CBAM-like schemes that track foreign policies.

2.4 Consumption-Based Emissions Accounting

A substantial literature in industrial ecology and climate policy argues that emissions responsibility should follow consumption. Peters (2008), Peters and Hertwich (2008), and Hertwich and Peters (2009) demonstrate that many high-income countries have consumption-based emissions far exceeding territorial emissions, due to outsourcing of carbon-intensive production. Jakob and Steckel (2014) and Jakob et al. (2014) emphasise that consumption-based frameworks better reflect equity considerations and the structure of global value chains.

This literature motivates OECD’s destination-basis approach: carbon charges should apply where goods are consumed, not where they are produced.

2.5 Consumption-Based Environmental Taxation and Carbon-Adjusted Tariff Systems

Consumption-based environmental taxation provides real-world examples of how environmental externalities can be incorporated into prices downstream.

Empirical evidence comes from:

- Nordic NO_x charges rebated according to performance benchmarks (Bruvoll and Larsen (2004));
- landfill taxes and waste-disposal levies in the UK (Elliott (2011));
- packaging, pesticide, and chemical taxes across Europe (OECD (2016));
- extended producer-responsibility schemes such as WEEE and battery directives (Tojo (2004));
- utility and retail energy levies including supplier-obligation schemes (Hood (2011)).

Recent research on consumption-based carbon pricing examines: retail electricity carbon charges (Vogt-Schilb and Hallegatte (2022)); point-of-sale carbon footprint fees (Summerfield (2019)); and platform-based carbon charges.

Parallel work considers ****carbon-adjusted tariffs****, a consumption-based mechanism implemented at the border: environmental border charges Lockwood and Whalley (2008); tariff refinement based on climate criteria Colares (2014); verified low-carbon tariff differentiation Électricité de France (2019); and embodied-emissions adjustments to customs schedules Moran and Wood (2020); Brandi (2023).

These literatures collectively show that consumption-based carbon taxation is administratively feasible, compatible with trade rules, and increasingly embedded in real-world environmental policy.

OECP builds directly on this foundation by combining a destination-based consumption charge with benchmark-based tariff adjustments.

2.6 Political Acceptability, Revenue Use, and Public Finance Analogies

Political-economy research shows that public support for carbon pricing depends heavily on how revenues are used. Metcalf (2009) and Metcalf (2017) emphasise transparent, equitable recycling. Parry and Williams (1999) analyse efficiency–equity trade-offs in revenue use, while Carbone et al. (2013) demonstrate the distributional impacts of alternative recycling mechanisms. Survey evidence synthesised in Klenert et al. (2018) shows that visible household transfers substantially increase acceptability.

The public-finance literature provides an important analogy: destination-based taxes are less distortionary than source-based taxes. Empirical studies on VAT adoption (Keen and Lockwood (2006) and Keen and Lockwood (2010)) show that VAT improved revenue stability and encouraged upward convergence because it taxes consumption rather than production. Theoretical work on destination-based corporate taxation (Auerbach et al. (2010)) shows similar advantages.

OECP applies these insights to carbon: it separates revenue generation (via the consumption charge) from incentive provision (via feebates), mirroring the advantages of moving from source-based to destination-based taxation.

3 The Two-Plus-One Functions of Carbon Pricing

A major contribution of this paper is clarifying that carbon pricing has **two established functions** and **one missing function** that current policy does not address.

3.1 Function 1: Transition (Economic Efficiency)

The first purpose of carbon pricing is to induce a shift from high-emission to low-emission technologies. This requires strong *relative* price signals, not necessarily high *average* cost increases. Feebates are ideal for this function.

3.2 Function 2: Polluter-Pays Responsibility

The second function is normative: entities that cause climate damage should pay for that damage. Under CCP, this function is combined with the transition function within a single instrument, creating tension between high prices for justice and low prices for feasibility. OECP separates these functions: the consumption charge handles responsibility, while the feebate handles incentives.

3.3 Function 3 (Missing): Remediation or Compensation

In moral philosophy and tort law, responsibility must be paired with:

- remediation of harm; or
- compensation to harmed parties.

Current systems do neither adequately. OECP, by relocating revenue generation to consumption, frees sufficient revenue for a future system of remediation and compensation. Designing such a system is beyond this paper’s scope but constitutes a natural extension of the OECP framework.

3.4 Why Combining Functions Breaks CCP

Under CCP:

- Transition requires high prices;
- Polluter-pays justice also requires high prices;
- Political feasibility constrains prices to low levels.

Combining functions within a single tax collapses under political pressure. OECP succeeds because it separates functions.

4 The Structural Constraints of Conventional Carbon Pricing

4.1 The Trade Constraint

Source-based carbon pricing raises both marginal and average production costs for domestic firms. Exports become less competitive, and imports—often from more carbon-intensive

producers—gain an advantage. Even with CBAM, domestic producers remain at a disadvantage in export markets. This creates structural leakage and a persistent race to the bottom in carbon prices.

4.2 The Political-Economy Constraint

Political feasibility under CCP requires keeping:

- industrial opposition manageable; and
- household price impacts tolerable.

A uniform carbon tax or ETS increases both. Empirically, feasible prices remain in the €30–€60/tCO₂ range, well below the level required for rapid structural transition.

4.3 The Transition Constraint

Primary sectors require capital turnover, not marginal adjustment. The carbon price required to shift investment in steelmaking, cement production, refining or power generation is far higher than the price politically feasible under CCP. Thus, CCP is structurally mismatched to the energy system’s primary–tertiary structure.

5 Introducing Open-Economy Carbon Pricing (OECF) Framework

OECF replaces Conventional Carbon Pricing (CCP) – which has a single, overloaded instrument – with a two-instrument system separating incentives (production feebate) from damage remediation (consumption based carbon pricing).

5.1 Production Feebate

Each sector sets a benchmark emissions intensity $\bar{e}$. A firm with emissions e_i receives a subsidy or pays a fee:

$$F_i = t(e_i - \bar{e}).$$

This preserves relative incentives while keeping sector-average costs unchanged. This eliminates export disadvantages and encourages technological switching.

5.2 Destination-Based Consumption Charge

All goods consumed domestically face a charge $t \times m_j$, where m_j is embodied emissions. Imports and domestic goods are treated symmetrically; exports are zero-rated. This mirrors VAT and restores trade neutrality.

5.3 Instrument Interaction

The feebate strengthens domestic decarbonisation competition. The consumption charge:

- raises revenue,
- applies polluter-pays principles,
- and eliminates trade distortions.

Together, they raise feasible incentive rates and avoid leakage.

5.4 Forward Links

Appendix A formalises the trade model underlying this logic. Appendix B develops the political feasibility model. Appendix C provides the full implementation details.

6 Accelerated Transition: The Primary–Tertiary Argument

Carbon pricing must align with the structure of the economy:

- primary sectors (fuels, steel, cement),
- secondary sectors (manufactured goods),
- tertiary sectors (services).

Most emissions originate upstream in primary sectors, yet downstream consumers drive demand. Conventional carbon pricing, by taxing only domestic production, fails to accelerate capital turnover in primary sectors because it:

- raises average costs (thus generating political resistance),
- but fails to provide strong relative incentives,

- and cannot reach required levels because of trade and voter constraints.

OECP resolves this:

- strong feebate incentives accelerate switching,
- destination-basis charges allow high effective carbon prices,
- export neutrality prevents industrial pushback,
- consumption charges align downstream demand with upstream transition.

This restores coherence between climate policy design and energy-system structure.

7 Implementation of OECP

OECP consists of three administratively simple elements:

1. a revenue-neutral production feebate,
2. a destination-based consumption carbon charge,
3. and border adjustments for imports and exports.

This section shows how these components can be implemented using existing institutional infrastructure, building on but simplifying the logic of the EU CBAM and universal VAT systems.

The overall objective is to ensure:

- strong decarbonisation incentives for producers,
- equal treatment of domestic and foreign goods,
- preserved export competitiveness,
- and politically sustainable revenue generation.

Implementation details requiring deeper technical specification are deferred to Appendix C (Tax Implementation), but summarised here for clarity.

7.1 Production Feebate: Incentives Without Cost Increases

In each sector, a benchmark emissions intensity $\bar{e}$ is chosen. Any producer i with emissions e_i receives a feebate:

$$F_i = t(e_i - \bar{e}).$$

The feebate has four critical properties:

1. It is approximately revenue-neutral at the sector level.
2. It creates strong relative incentives to decarbonise.
3. It does not raise average production costs.
4. It leaves export prices unchanged.

This is the core instrument that resolves CCP's competitiveness constraint. Because average costs do not increase, firms do not lobby against high incentive rates. Producers remain competitive in global markets even when t is high.

7.2 Consumption Carbon Charge: Revenue and Responsibility

All goods consumed domestically—regardless of origin—bear a carbon charge:

$$C_j = t \cdot m_j,$$

where m_j is embodied emissions in product j .

This charge:

- applies identically to imports and domestic goods,
- ensures polluter-pays responsibility,
- raises revenue without imposing costs on exports,
- is compatible with WTO norms under GATT Article II:2(a),
- and mirrors VAT systems, which tax consumption rather than production.

The consumption charge is the locus of revenue generation within OECP. It replaces the revenue-raising function of CCP and allows future work (Section ??) to allocate revenue for remediation and compensation.

7.3 Import Adjustments: A Simplified CBAM

Imports are charged based on embodied emissions. The system is intentionally simple:

1. Importer declares either verified emissions or accepts a conservative default.
2. Customs applies $t \cdot m_j$, using:
 - a reference intensity (e.g. China or global high-bound),
 - a country-specific adjustment,
 - optional producer-specific data for large firms.

This structure avoids CBAM’s complexity:

- no tracking of foreign free allocations,
- no foreign ETS equivalence calculations,
- and no need for the EU-style “carbon certificate market.”

Implementation can follow:

- **category-level intensities** for EITE sectors (steel, cement, aluminium),
- **value-based intensities** for general goods,
- **physical-unit intensities** for bulk commodities (see Appendix C).

7.4 Export Treatment: VAT-Style Zero Rating

Exports face *no* consumption charge. Producers still participate in the feebate but pay no absolute carbon tax on exports.

Thus exports remain competitive, and domestic producers are not harmed by high carbon incentive rates. This preserves neutrality across jurisdictions and ensures WTO compatibility.

7.5 Common External Tariff Variant

In customs unions, a **Carbon-Adjusted Common External Tariff (CACET)** can simplify implementation:

- A base tariff applies to all countries.

- A carbon adjustment is added:

$$\text{CAT}_j = t \cdot (m_j - m_{\text{benchmark}}).$$

- Countries with verified low-carbon production receive reduced charges.

This approach is particularly suitable for the EU, ASEAN, EAC, and MERCOSUR.

7.6 International Incentives: Why OECP Encourages a Race to the Top

OECP changes the international incentive structure by rewarding lower-carbon production regardless of jurisdiction.

Under CCP: A country that decarbonises its industry faces higher domestic costs and no advantage for exports. Thus the incentive is weak.

Under OECP: A country that decarbonises:

- receives lower import charges in foreign markets (as foreign consumers pay less),
- secures competitive gains for its domestic producers,
- and retains strong feebate incentives internally.

This restores the same logic that made VAT universally adoptable: destination-basis taxes remove competitiveness harms and induce upward convergence.

7.7 Transfers and Feebate Gaps

A subtle issue arises when the feebate incentive becomes strong enough to trigger switching to a technology that is initially more expensive. In this case:

- below the switching threshold, feebates generate surplus revenue;
- above the threshold, feebates may become revenue-negative.

The gap can be financed by:

- revenue from the consumption charge, or
- international transfers (if used in developing countries),

- multilateral development bank facilities,
- climate finance arrangements.

In a simple two-technology model:

- if clean technology is 10% more expensive,
- and if the effective feebate rate reduces average cost increases to 2–3%,

then the transfer requirement is small relative to the total revenue potential of the consumption charge.

Appendix A provides a formal derivation of this result.

8 Link to Appendices

Part III provides:

- Appendix A: A formalised economic model of trade and transition.
- Appendix B: A detailed political-economy model showing the feasibility frontier and how OECP expands it.
- Appendix C: A complete “Tax Implementation” appendix covering import charges, country-specific adjustment, product granularity, and export zero-rating.

A Appendix A: Economic Model of Trade, Transition, and OECP

This appendix presents a minimal open-economy model showing:

1. why conventional carbon pricing (CCP) creates trade distortions,
2. why feebates preserve competitiveness,
3. how destination-based charges eliminate leakage,
4. and how OECP raises feasible incentive rates.

The model favours simplicity over formal complexity and is intended to provide clear intuition for the results used in the main text.

A.1 A.1 Basic Setup

Two countries: Home (H) and Foreign (F). One homogeneous good is produced using one of two technologies:

- *Dirty technology (D)*: cost c_D , emissions e_D .
- *Clean technology (C)*: cost $c_C > c_D$, emissions $e_C < e_D$.

Demand in Home is $Q_H(p)$; demand in Foreign is $Q_F(p)$.

Foreign imposes no carbon constraints unless otherwise stated.

Trade is frictionless, and firms are price-takers.

A.2 A.2 Conventional Carbon Pricing

Under a source-based carbon tax t , Home producers face cost:

$$p_i^{\text{CCP}} = c_i + te_i.$$

Thus:

- Clean production becomes more competitive relative to dirty production.
- **But** both clean and dirty production become less competitive relative to Foreign producers.

Exports shrink; imports expand. Leakage occurs if the increase in Foreign output exceeds the decline in Home output.

The equilibrium carbon price under CCP is limited by:

$$t \leq t^{\max} = \min(\tau_I, \tau_V),$$

where τ_I is the industry tolerance threshold, and τ_V the voter threshold.

A.3 A.3 OECP Feebate

Under OECP, producers face a feebate:

$$F_i = t(e_i - \bar{e}),$$

yielding a net cost:

$$p_i^{\text{FB}} = c_i + t(e_i - \bar{e}).$$

If the benchmark $\bar{e}$ is chosen such that:

$$\sum_i F_i = 0,$$

then average production costs *do not change*. Thus:

- Home producers do not suffer export losses.
- Foreign producers gain no competitive advantage.
- Strong relative incentives remain for clean producers.

A.4 A.4 OECP Consumption Charge

Home applies a consumption-based carbon charge:

$$C_j = t \cdot m_j,$$

to all *goods consumed domestically*, where m_j is embodied emissions.

Thus:

- Imports pay $t \cdot m_j$ at the border.
- Domestic goods pay $t \cdot m_j$ at point of sale.
- Exports pay no charge (zero-rated).

A.5 A.5 Results

Result 1: No leakage Because all consumption faces the same charge:

$$p^{\text{dom}}(j) = p^{\text{imp}}(j),$$

imports do not displace domestic production.

Result 2: Preservation of export competitiveness Because feebates do not raise average costs:

$$p_i^{\text{export}} = c_i + t(e_i - \bar{e}) \approx c_i.$$

Exports remain unaffected.

Result 3: Higher feasible incentive rate Since neither households nor industry face large average-cost increases:

$$t_{\text{OECP}} > t_{\text{CCP}}.$$

Result 4: Accelerated switching The switching condition under OECP becomes:

$$c_C - c_D < t(e_D - e_C),$$

which can be satisfied at moderate values of t , because producers face high relative incentives but no average-cost increases.

B Appendix B: Political Economy Model

This appendix develops a simple but more detailed model of political feasibility, using two constraints:

1. industry tolerance (τ_I),
2. voter tolerance (τ_V).

CCP faces both constraints; OECP reshapes them.

B.1 B.1 Utility and Cost Functions

Let t be the carbon incentive rate.

For industry, the relevant cost under CCP is:

$$\Delta c_I(t) = t \cdot \bar{e}_{\text{industry}},$$

which increases average costs and reduces profits:

$$\pi_I(t) = \pi_0 - Q \cdot \Delta c_I(t).$$

Industry accepts CCP if:

$$\Delta c_I(t) \leq \tau_I.$$

For households, the relevant cost under CCP is:

$$\Delta c_V(t) = \alpha t,$$

where α reflects the share of consumption affected and salience of energy price changes. Households accept CCP if:

$$\Delta c_V(t) \leq \tau_V.$$

Thus:

$$t_{\text{CCP}}^{\max} = \min(\tau_I / \bar{e}_{\text{industry}}, \tau_V / \alpha).$$

Empirically, both are low.

B.2 B.2 How OECP Changes the Constraints

Industry: Under OECP:

$$\Delta c_I^{\text{OECP}}(t) \approx 0.$$

The feebate preserves average costs:

$$p_i^{\text{FB}} \approx c_i,$$

so:

$$\tau_I^{\text{OECP}} \rightarrow \infty \quad (\text{no meaningful constraint}).$$

Households: The consumption charge raises prices for some goods, but revenue can be recycled back to households through:

- dividends,
- reduced labour taxes,
- rebates to low-income households,

- support for energy efficiency.

Thus the net household cost is:

$$\Delta c_V^{\text{OECF}}(t) = \alpha t - R(t),$$

where $R(t)$ is the per-household rebate.

If $R(t)$ is designed to grow with t , then:

$$\Delta c_V^{\text{OECF}}(t) \approx 0.$$

Thus:

$$t_{\text{OECF}}^{\max} \gg t_{\text{CCP}}^{\max}.$$

B.3 B.3 Feasible Policy Region

The feasible region in (t, rebate) space expands dramatically:

$$\text{Feasible}_{\text{OECF}} = \{(t, R) : \Delta c_I^{\text{OECF}}(t) \approx 0, \Delta c_V^{\text{OECF}}(t) \leq \tau_V\}.$$

In graphical terms:

- The industry constraint disappears.
- The voter constraint becomes a rising function of R .

Thus OECF opens political space for high incentive rates combined with household protection.

C Appendix C: Tax Implementation

This appendix provides a full specification of the tax implementation system summarised in Part II.

C.1 C.1 Overview of the System

Imports, domestic production, and exports are treated as follows:

1. **Production feebate:** revenue-neutral, applied to all domestic producers.

2. **Consumption charge:** applied to all goods consumed domestically.
3. **Import adjustments:** based on embodied emissions.
4. **Export zero-rating:** no consumption charge on exports.

C.2 C.2 Country-Level Adjustment

Each exporting country k receives a default intensity:

$$m_{jk}^{\text{default}} = m_j^{\text{ref}} + \delta_k,$$

where:

- m_j^{ref} is a high-bound reference intensity (e.g. China),
- δ_k is an adjustment for country k 's sectoral emissions profile.

Large firms can optionally submit producer-specific data m_{jk}^{firm} .

C.3 C.3 Product Granularity Options

Three methods:

- (1) **Category-level** : HS-code based categories with default intensities.
- (2) **Physical-unit based** : tCO₂ per tonne of steel, cement, aluminium, etc.
- (3) **Value-based** : tCO₂ per \$ of imported product.

Recommendation: Hybrid approach:

- category or physical-unit for EITE sectors,
- value-based for general goods.

C.4 C.4 Export Rebates

Exports are zero-rated. Producers still receive the feebate but pay no consumption charge.

Optional variants:

- input-level rebates (VAT-style),

- benchmark-based rebates,
- sector-level fixed rebates.

Recommended: **Pure zero-rating + feebate retention** (simplest, WTO-compatible).

C.5 C.5 Administrative Steps

1. Importer submits emissions data (verified or default).
2. Customs applies the consumption charge.
3. Domestic producers apply feebate and charge consumption tax.
4. Exporters zero-rate the consumption charge.
5. Government recycles revenue to households or remediation functions.

D Appendix D: Notes on WTO Compatibility

Although not elaborated in the main text, OECP is consistent with:

- GATT Article II:2(a): border adjustments for internal taxes,
- GATT Article III: national treatment,
- The SCM Agreement: feebates are non-discriminatory,
- Established VAT jurisprudence (e.g. EU CJEU decisions).

A full WTO-law analysis is left for future work.